

with elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure

- Rare jaundice, agranulocytosis
- Rare seizures
- As a class, antipsychotics are associated with an increased risk of death and cerebrovascular events in elderly patients with dementia; not approved for treatment of dementia-related psychosis



Weight Gain

- Reported but not expected



Sedation

- Many experience and/or can be significant in amount
- Sedation is usually transient

What to Do About Side Effects

- Wait
- Wait
- Wait
- For drug-induced parkinsonism, add an anticholinergic agent
- Beta blockers, benzodiazepines, or 5HT2A antagonists (e.g., mirtazapine, cyproheptadine) may reduce akathisia
- Reduce the dose

- For sedation, give at night
- Switch to an atypical antipsychotic
- Weight loss, exercise programs, and medical management for high BMIs, diabetes, dyslipidemia
- Metformin may help prevent or reverse antipsychotic-induced weight gain

Best Augmenting Agents for Side Effects

- Benztropine, trihexyphenidyl, or amantadine for drug-induced parkinsonism
- Beta blockers, benzodiazepines, or 5HT_{2A} antagonists (e.g., mirtazapine, cyproheptadine) for akathisia
- Valbenazine or deutetrabenazine for tardive dyskinesia
- Many side effects cannot be improved with an augmenting agent

Dosing And Use

Usual Dosage Range

- Oral: psychosis: 15–20 mg/day

Dosage Forms

- Tablet 1 mg, 2 mg, 5 mg, 10 mg
- Vial 2 mg/mL (discontinued in USA)

- Concentrate 10 mg/mL (discontinued in USA)

How to Dose

- Psychosis: oral: initial 2–5 mg twice a day; increase gradually over 2–3 weeks
- Psychosis: intramuscular: 1–2 mg every 4–6 hours; generally do not exceed 6 mg/day
- Anxiety: initial 1–2 mg/day; maximum 6 mg/day



Dosing Tips

- * Use only low doses and short-term for anxiety because trifluoperazine is now a second-line treatment and has the risk of tardive dyskinesia
- Concentrate contains sulfites that may cause allergic reactions, particularly in patients with asthma
- Many patients can be dosed once a day
- Rather than raise the dose above normal dosing in acutely agitated patients requiring acute antipsychotic actions, consider short-term (one dose or more) augmentation with a benzodiazepine or another antipsychotic, especially intramuscularly
- Rather than raise the dose above normal dosing in partial responders, consider augmentation with a mood-stabilizing anticonvulsant, such as valproate, topiramate, or lamotrigine

- Children and elderly should generally be dosed at the lower end of the dosage spectrum
- Treatment should be suspended if absolute neutrophil count falls below $1000/\text{mm}^3$

Overdose

- Drug-induced parkinsonism, sedation, seizures, coma, hypotension, respiratory depression

Long-Term Use

- Should periodically reevaluate long-term usefulness in individual patients, but treatment may need to continue for many years
- Not intended to treat anxiety long-term (i.e., longer than 12 weeks)

Habit Forming

- No

How to Stop

- Slow down-titration of oral formulation (over 6–8 weeks), especially when simultaneously beginning a new antipsychotic while switching (i.e., cross-titration)
- Rapid oral discontinuation may lead to rebound psychosis and worsening of symptoms

- If antiparkinson agents are being used, they should be continued for a few weeks after trifluoperazine is discontinued

Pharmacokinetics

- Mean elimination half-life approximately 12.5 hours



Drug Interactions

- May decrease the effects of levodopa, dopamine agonists
- May increase the effects of antihypertensive drugs except for guanethidine, whose antihypertensive actions trifluoperazine may antagonize
- Additive effects may occur if used with CNS depressants
- Alcohol and diuretics may increase the risk of hypotension; epinephrine may lower blood pressure
- Phenothiazines may reduce effects of anticoagulants
- Some patients taking a neuroleptic and lithium have developed an encephalopathic syndrome similar to neuroleptic malignant syndrome
- If used with propranolol, plasma levels of both drugs may rise



Other Warnings/Precautions

- If signs of neuroleptic malignant syndrome develop, treatment should be immediately discontinued

- Use cautiously in patients with alcohol withdrawal or convulsive disorders because of possible lowering of seizure threshold
- Use with caution in patients with respiratory disorders, glaucoma, or urinary retention
- Avoid undue exposure to sunlight
- Avoid extreme heat exposure
- Antiemetic effect may mask signs of other disorders or overdose; suppression of cough reflex may cause asphyxia
- Do not use epinephrine in event of overdose as interaction with some pressor agents may lower blood pressure
- Use only with caution if at all in Parkinson's disease or Lewy body dementia
- Antipsychotics may cause somnolence, postural hypotension, motor and sensory instability, which may lead to falls; for patients with diseases, conditions, or medications that could exacerbate these effects, complete fall risk assessments when initiating antipsychotic treatment, and recurrently for patients on long-term antipsychotic therapy
- As with any antipsychotic, use with caution in patients with history of seizures

Do Not Use

- If patient is in a comatose state or has CNS depression

- If there is the presence of blood dyscrasias, bone marrow depression, or liver disease
- If there is a proven allergy to trifluoperazine
- If there is a known sensitivity to any phenothiazine

Special Populations

Renal Impairment

- Use with caution

Hepatic Impairment

- Not recommended for use

Cardiac Impairment

- Dose should be lowered
- Do not use parenteral administration unless necessary

Elderly

- Lower doses should be used and patient should be monitored closely
- Although conventional antipsychotics are commonly used for behavioral disturbances in dementia, no agent has been approved for treatment of elderly patients with behavioral symptoms of dementia such as agitation

- Elderly patients with dementia-related psychosis treated with antipsychotics are at an increased risk of death compared to placebo, and also have an increased risk of cerebrovascular events

Children and Adolescents

- Not recommended for use in children under age 6
- Children should be closely monitored when taking trifluoperazine
- Oral: initial 1 mg; increase gradually; maximum 15 mg/day except in older children with severe symptoms
- Intramuscular: 1 mg once or twice a day
- Generally consider second-line after atypical antipsychotics

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- There is a risk of abnormal muscle movements and withdrawal symptoms in newborns whose mothers took an antipsychotic during the third trimester; symptoms may include agitation, abnormally

increased or decreased muscle tone, tremor, sleepiness, severe difficulty breathing, and difficulty feeding

- Reports of drug-induced parkinsonism, jaundice, hyperreflexia, hyporeflexia in infants whose mothers took a phenothiazine during pregnancy
- Trifluoperazine should only be used during pregnancy if clearly needed
- Psychotic symptoms may worsen during pregnancy and some form of treatment may be necessary
- Atypical antipsychotics may be preferable to conventional antipsychotics or anticonvulsant mood stabilizers if treatment is required during pregnancy

Breast Feeding

- Some drug is found in mother's breast milk
- * Recommended either to discontinue drug or bottle feed

The Art Of Psychopharmacology

Potential Advantages

- Intramuscular formulation for emergency use

Potential Disadvantages

- Patients with tardive dyskinesia

- Children
- Elderly

Primary Target Symptoms

- Positive symptoms of psychosis
- Motor and autonomic hyperactivity
- Violent or aggressive behavior

Pearls

- Trifluoperazine is a higher potency phenothiazine
- * Although not systematically studied, may cause less weight gain than other antipsychotics
- Less risk of sedation and orthostatic hypotension but greater risk of drug-induced parkinsonism than with low-potency
- Patients have very similar antipsychotic responses to any conventional antipsychotic, which is different from atypical antipsychotics where antipsychotic responses of individual patients can occasionally vary greatly from one atypical antipsychotic to another
- Patients with inadequate responses to atypical antipsychotics may benefit from a trial of augmentation with a conventional antipsychotic such as trifluoperazine or from switching to a conventional antipsychotic such as trifluoperazine

- However, long-term polypharmacy with a combination of a conventional antipsychotic such as trifluoperazine with an atypical antipsychotic may combine their side effects without clearly augmenting the efficacy of either
- For treatment-resistant patients, especially those with impulsivity, aggression, violence, and self-harm, long-term polypharmacy with 2 atypical antipsychotics or with 1 atypical antipsychotic and 1 conventional antipsychotic may be useful or even necessary while closely monitoring
- In such cases, it may be beneficial to combine 1 depot antipsychotic with 1 oral antipsychotic
- Although a frequent practice by some prescribers, adding 2 conventional antipsychotics together has little rationale and may reduce tolerability without clearly enhancing efficacy

Suggested Reading

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Huhn M , Nikolakopoulou A , Schneider-Thoma J , et al. Comparative efficacy and tolerability of 32 oral antipsychotics for the acute treatment of

adults with multi-episode schizophrenia: a systematic review and network meta-analysis . *Lancet* 2019 ;**394** :939–51 .

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Kiloh LG , Williams SE , Grant DA , Whetton PS . A double-blind comparative trial of loxapine and trifluoperazine in acute and chronic schizophrenic patients . *J Int Med Res* 1976 ;**4** :441 –8 .

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Trihexyphenidyl

Therapeutics

Brands

- Artane
- *see index for additional brand names*

Generic?

Yes

Class

- Antiparkinson agent; anticholinergic

Commonly Prescribed for

(bold for FDA approved)

- **Extrapyramidal disorders (drug-induced parkinsonism)**
- **Parkinsonism**
- Idiopathic generalized dystonia
- Focal dystonias
- Dopa-responsive dystonia

How the Drug Works

- Diminishes the excess acetylcholine activity caused by removal of dopamine inhibition when dopamine receptors are blocked
- May also inhibit the reuptake and storage of dopamine at central dopamine receptors, prolonging dopamine action

How Long Until It Works

- For drug-induced parkinsonism and in Parkinson's disease, onset of action can be within minutes or hours

If It Works

- Reduces motor side effects
- Does not lessen the ability of antipsychotics to cause tardive dyskinesia

If It Doesn't Work

- For drug-induced parkinsonism, increase to highest tolerated dose
- Consider switching to benztropine, diphenhydramine, or a benzodiazepine
- Disorders that develop after prolonged antipsychotic use may not respond to treatment
- Consider discontinuing the agent that precipitated the parkinsonism



Best Augmenting Combos for Partial Response or Treatment Resistance

- If ineffective, switch to another agent rather than augment
- Trihexyphenidyl itself is an augmenting agent to antipsychotics

Tests

- None for healthy individuals

Side Effects

How Drug Causes Side Effects

- Prevents the action of acetylcholine on muscarinic receptors

Notable Side Effects

- Dry mouth, blurred vision, diplopia
- Confusion, hallucinations
- Constipation, nausea, vomiting
- Dilation of colon
- Erectile dysfunction



Life-Threatening or Dangerous Side Effects

- Angle-closure glaucoma
- Heat stroke, especially in elderly patients
- Tachycardia, cardiac arrhythmias, hypotension

- Urinary retention
- Anticholinergic agents such as trihexyphenidyl can exacerbate or unmask tardive dyskinesia

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount

What to Do About Side Effects

- For confusion or hallucinations, discontinue use
- For sedation, lower the dose and/or take the entire dose at night
- For dry mouth, chew gum or drink water
- For urinary retention, obtain a urological evaluation; may need to discontinue use

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing And Use

Usual Dosage Range

- Drug-induced parkinsonism: 5–15 mg/day
- Parkinsonism: 6–15 mg/day

Dosage Forms

- Tablet 2 mg, 5 mg
- Injection 2 mg/5 mL

How to Dose

- Drug-induced parkinsonism: 5–15 mg/day; assess effect and increase dose empirically as tolerated; total daily dose varies from patient to patient
- Parkinsonism (oral): initial 1 mg/day; after 3 days can increase the dose in 2 mg increments every 3–5 days as tolerated until clinical effect is reached; total daily dose should be divided into 3 doses and given with meals



Dosing Tips

- If drug-induced parkinsonism occurs soon after initiation of a neuroleptic drug, it is likely to be transient; thus, attempt to withdraw trihexyphenidyl after 1–2 weeks to determine if still needed
- To achieve more rapid relief, temporarily lower the dose of the offending agent (phenothiazine, thioxanthene, or butyrophenone)

when starting trihexyphenidyl

- Taking trihexyphenidyl with meals can reduce side effects
- In Parkinson's disease, the usual dose is 6–10 mg/day (divided into 3 doses) for idiopathic Parkinson's disease and 12–15 mg/day (divided into 3 or 4 doses) for post-encephalitic Parkinson's disease

Overdose

- Circulatory collapse, cardiac arrest, respiratory depression or arrest, CNS depression or stimulation, psychosis, shock, coma, seizure, ataxia, combativeness, anhidrosis and hyperthermia, fever, dysphagia, decreased bowel sounds, sluggish pupils

Long-Term Use

- Safe
- Effectiveness may decrease over time (years) and side effects such as sedation and cognitive impairment may worsen

Habit Forming

- No

How to Stop

- Taper not necessary

Pharmacokinetics

- Half-life 6–10 hours; time to peak effect is 1–1.3 hours
- Metabolism is not well understood



Drug Interactions

- Use with amantadine may increase side effects
- Trihexyphenidyl and all other anticholinergic agents may increase serum levels and effects of digoxin
- Can lower concentration of haloperidol and other phenothiazines, causing worsening of schizophrenia symptoms
- Can decrease gastric motility, resulting in increased gastric deactivation of levodopa and reduction in efficacy



Other Warnings/Precautions

- Use with caution in hot weather, as trihexyphenidyl may increase susceptibility to heat stroke
- Anticholinergic agents have additive effects when used with drugs of abuse such as cannabinoids, barbiturates, opioids, and alcohol

Do Not Use

- In patients with glaucoma, particularly angle-closure glaucoma
- In patients with pyloric or duodenal obstruction, stenosing peptic ulcers, prostate hypertrophy or bladder neck obstructions, achalasia, or megacolon

- If there is a proven allergy to trihexyphenidyl

Special Populations

Renal Impairment

- No known effects, but use with caution

Hepatic Impairment

- No known effects, but use with caution

Cardiac Impairment

- Use with caution in patients with known arrhythmias, especially tachycardia

Elderly

- Use with caution
- Elderly patients may be more susceptible to side effects



Children and Adolescents

- Do not use in children ages 3 and younger
- Generalized dystonias may respond to anticholinergic treatment, and young patients usually tolerate the medication better than the elderly
- Usual dose is 0.05 mg/kg once or twice daily



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women

Breast Feeding

- Unknown if trihexyphenidyl is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed unless the potential benefit to the mother justifies the potential risk to the child
- Infants of women who choose to breast feed while on trihexyphenidyl should be monitored for possible adverse effects

The Art Of Psychopharmacology

Potential Advantages

- Drug-induced parkinsonism
- Generalized dystonias (well tolerated in younger age groups)

Potential Disadvantages

- Patients with long-standing drug-induced parkinsonism may not respond to treatment
- Multiple dose-dependent side effects may limit use

Primary Target Symptoms

- Tremor, akinesia, rigidity, drooling, dystonia

Pearls

- Often abused in correctional settings for its euphoric and sedative/hypnotic action, especially at high doses
- Although any anticholinergic agent is potentially abusable, especially in a correctional setting, trihexyphenidyl may be more abusable than others, presumably due to its dopamine-enhancing actions
- Useful adjunct in younger Parkinson's patients with tremor
- Useful in the treatment of post-encephalitic Parkinson's disease and for drug-induced parkinsonism, but not for tardive dyskinesias
- Post-encephalitic Parkinson's patients usually tolerate higher doses better than idiopathic Parkinson's patients
- Generalized dystonias are more likely to benefit from anticholinergic therapy than focal dystonias; trihexyphenidyl is used more commonly than benztropine
- Sedation limits use, especially in older patients

- Patients with mental impairment do poorly
- Dystonias related to cerebral palsy, head injuries, and stroke may improve with trihexyphenidyl, especially in younger, cognitively normal patients
- Schizophrenia patients may abuse trihexyphenidyl and other anticholinergic medications to relieve negative symptoms, for a stimulant effect, or to improve symptoms of drug-induced parkinsonism
- Can cause cognitive side effects with chronic use, so periodic trials of discontinuation may be useful to justify continuous use, especially in institutional settings as adjunct to antipsychotics

Suggested Reading

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Sanger TD , Bastian A , Brunstrom J , et al. Prospective open-label clinical trial of trihexyphenidyl in children with secondary dystonia due to cerebral palsy . *J Child Neurol* 2007 ;**22** (5):530–7 .

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Zemishlany Z , Aizenberg D , Weiner Z , Weizman A. Trihexyphenidyl (Artane) abuse in schizophrenic patients . *Int Clin Psychopharmacol* 1996 ;**11** (3):199 –202 .

Triiodothyronine

Therapeutics

Brands

- Cytomel
- *see index for additional brand names*

Generic?

Yes

Class

- Synthetic hormone; antidepressant adjunct

Commonly Prescribed for

(bold for FDA approved)

- **Replacement or supplemental therapy in patients with hypothyroidism (except transient hypothyroidism during the recovery phase of subacute thyroiditis)**
- **Pituitary thyroid-stimulating hormone (TSH) suppressant in the treatment or prevention of various types of euthyroid goiters**
- Major depressive disorder (adjunct)

How the Drug Works